

J. Backes^{a,c}, A. Tsanda^{a,b}, W. Renz^c, E. Schöll^d and T. Knopp^{a,b}

^aInstitute for Biomedical Imaging, Hamburg University of Technology (TUHH), Hamburg, Germany

^bSection for Biomedical Imaging, University Medical Center Hamburg-Eppendorf (UKE), Hamburg, Germany

^cResearch Group Distributed Adaptive Systems, Hamburg University of Applied Sciences (HAW), Hamburg, Germany

^dInstitute for theoretical Physics, Technical University Berlin (TUB), Berlin, Germany

Physiological Network Model

The "Physiological Network Model", introduced in [1] and mathematically defined in Eqs. 1-4, represents the interaction between parenchymal cells (functional organ cells; opposed to stroma the structural cells) denoted φ_i^1 , and immune cells, denoted φ_i^2 , using a two layer partly adaptive oscillator network, with $i = 1, \dots, N$, where $\kappa_{ij}^{1/2}$ are adaptive coupling weights.

$$\begin{aligned} \dot{\varphi}_i^1 &= -\frac{1}{N} \sum_{j=1}^N ((1 + \kappa_{ij}^1) \sin(\varphi_i^1 - \varphi_j^1 - 0.28\pi)) - \sigma \sin(\varphi_i^1 - \varphi_i^2) \quad (1) \\ \dot{\kappa}_{ij}^1 &= -0.03 (\kappa_{ij}^1 + \sin(\varphi_i^1 - \varphi_j^1 - \beta)) \quad (2) \\ \dot{\varphi}_i^2 &= -\frac{1}{N} \sum_{j=1}^N (\kappa_{ij}^2 \sin(\varphi_i^2 - \varphi_j^2 + -0.28\pi)) - \sigma \sin(\varphi_i^2 - \varphi_i^1) \quad (3) \\ \dot{\kappa}_{ij}^2 &= -0.3 (\kappa_{ij}^2 + \sin(\varphi_i^2 - \varphi_j^2 - \beta)) \quad (4) \end{aligned}$$

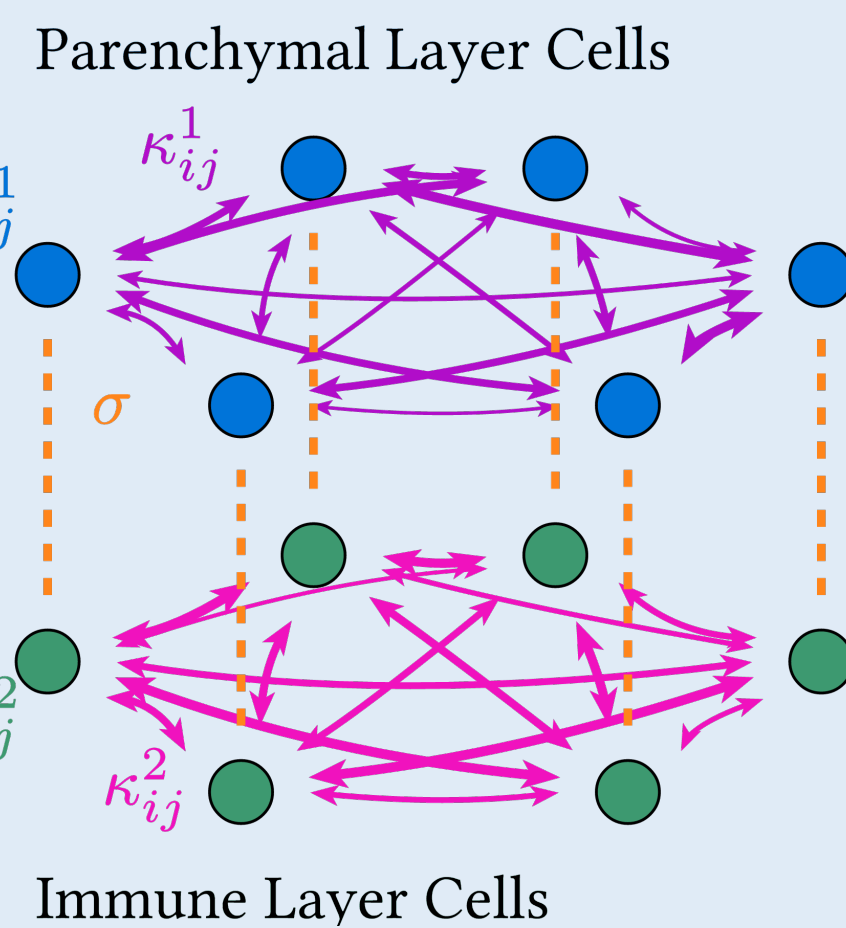


Figure 1: Physiological Network Model

Different stable synchronization patterns can occur based on the parameterization. It is hypothesized that synchronized frequencies in the parenchymal layer represent a healthy patient, whereas desynchronized states most likely capture pathological conditions [2]. The amount of desynchronisation is expressed by the *ensemble average of the standard deviation of the mean phase velocities*, defined by:

$$s^1 = \frac{1}{M} \sum \sqrt{\frac{1}{N} \sum_j (\langle \dot{\varphi}_j^1 \rangle - \bar{\omega}^1)^2} \quad (5)$$

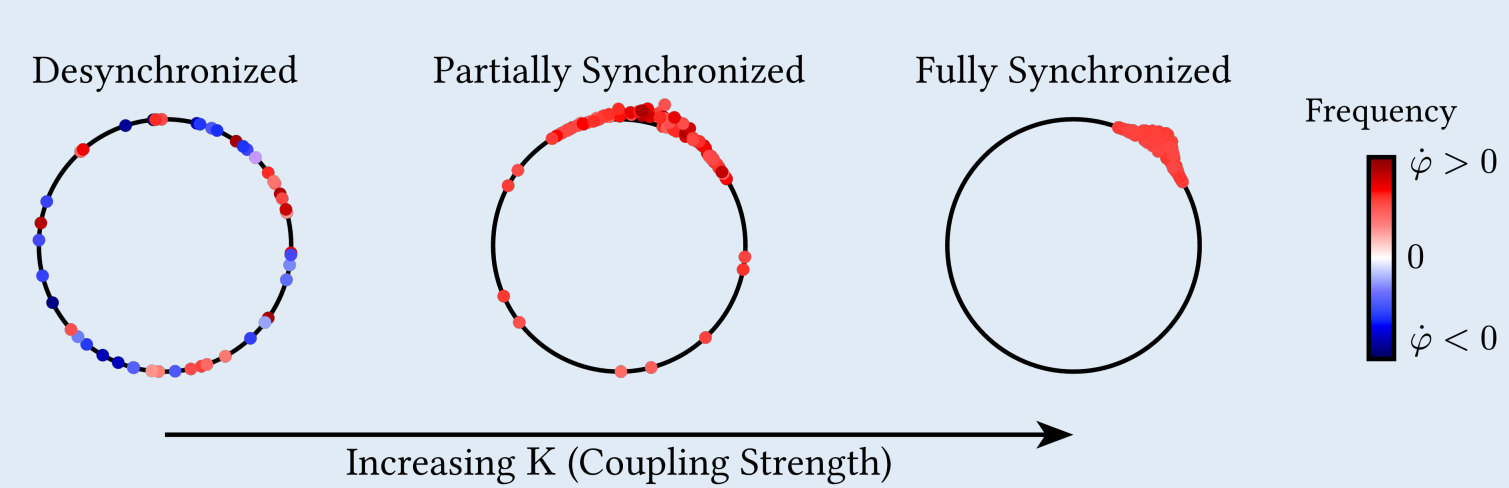


Figure 2: Synchronization behaviour based on coupling strength

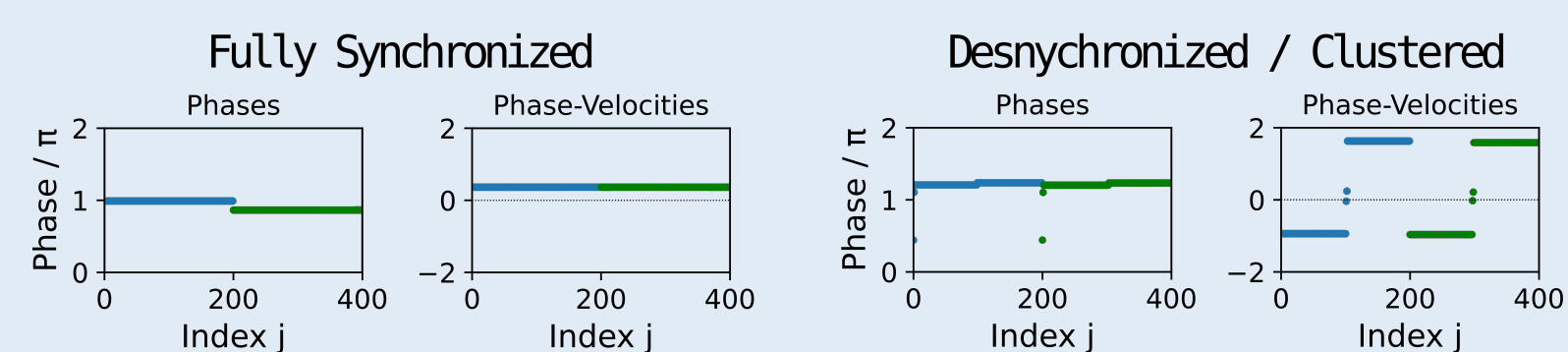


Figure 3: Stable States; Snapshots of Parenchymal Cells (blue) and Immune Cells (green)

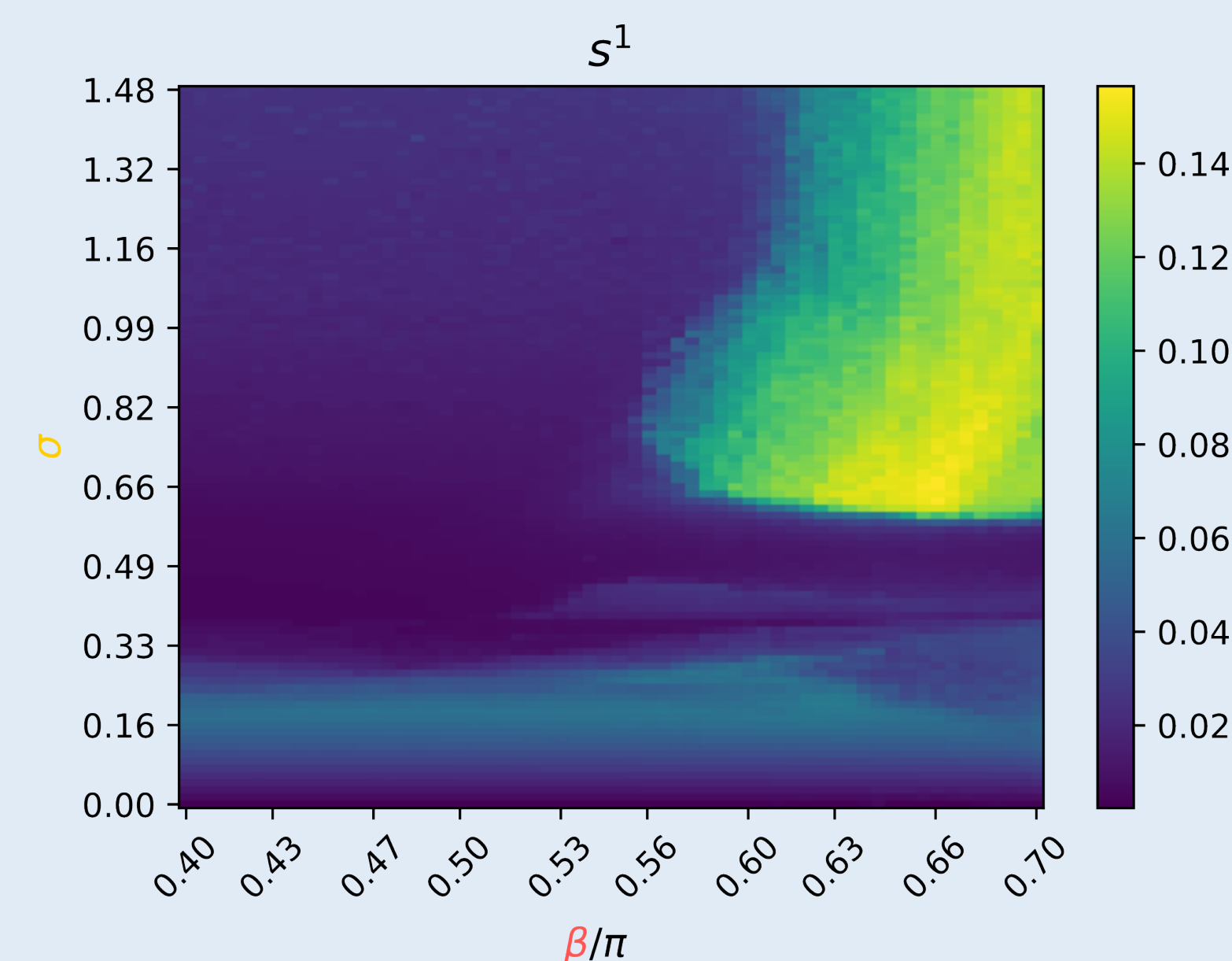


Figure 4: Parameter Space; Brighter colors indicate larger desynchronization (corresponding to worse organ functionality)

Synchronization behavior can be modulated by the two parameters β (biological age) and σ (immun-organ interaction strength).

Sepsis

Critical condition defined by:

1. (Suspected) Infection
 2. Acute Organ Failure (increase in SOFA score ≥ 2)
- ~ **Sequential Organ Failure Assessment** [3]
- Assesses failure of 6 organ systems
 - Higher score (0-24) means worse condition

Latent Dynamics Model

A machine-learning pipeline that learns to project high-dimension patient data μ_t into the low-dimensional parameter space. The generated trajectories through this space allow the prediction of critical organ system states.

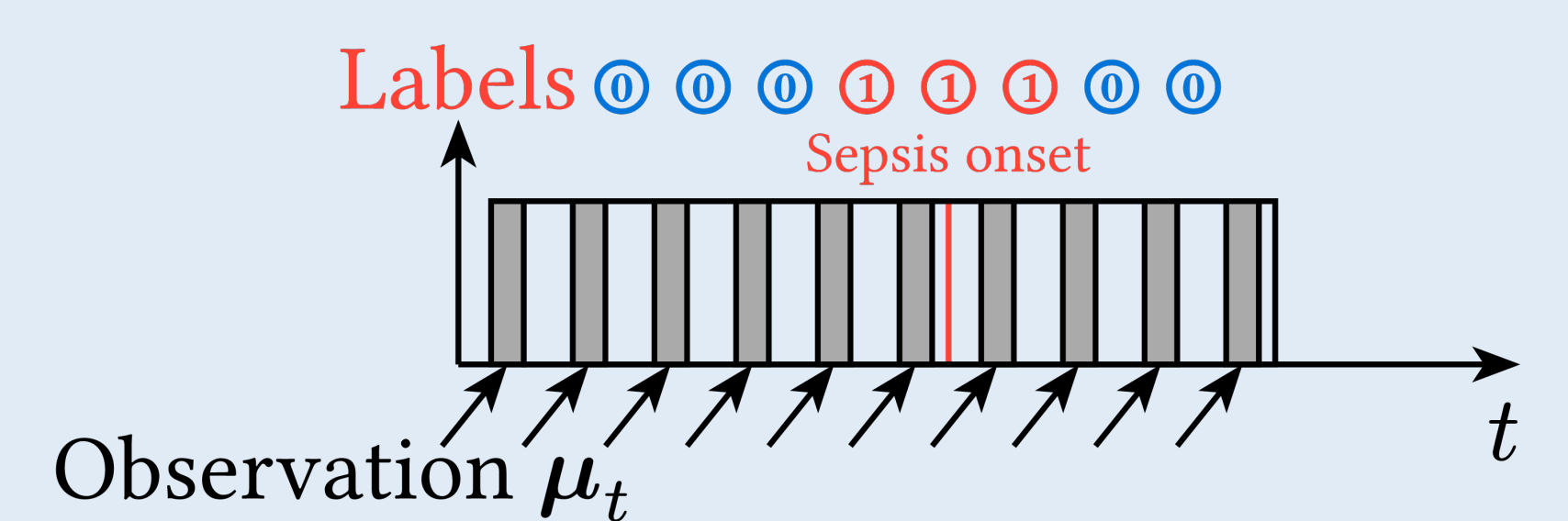


Figure 5: Online Prediction Methodology; Update Sepsis risk based on new observations

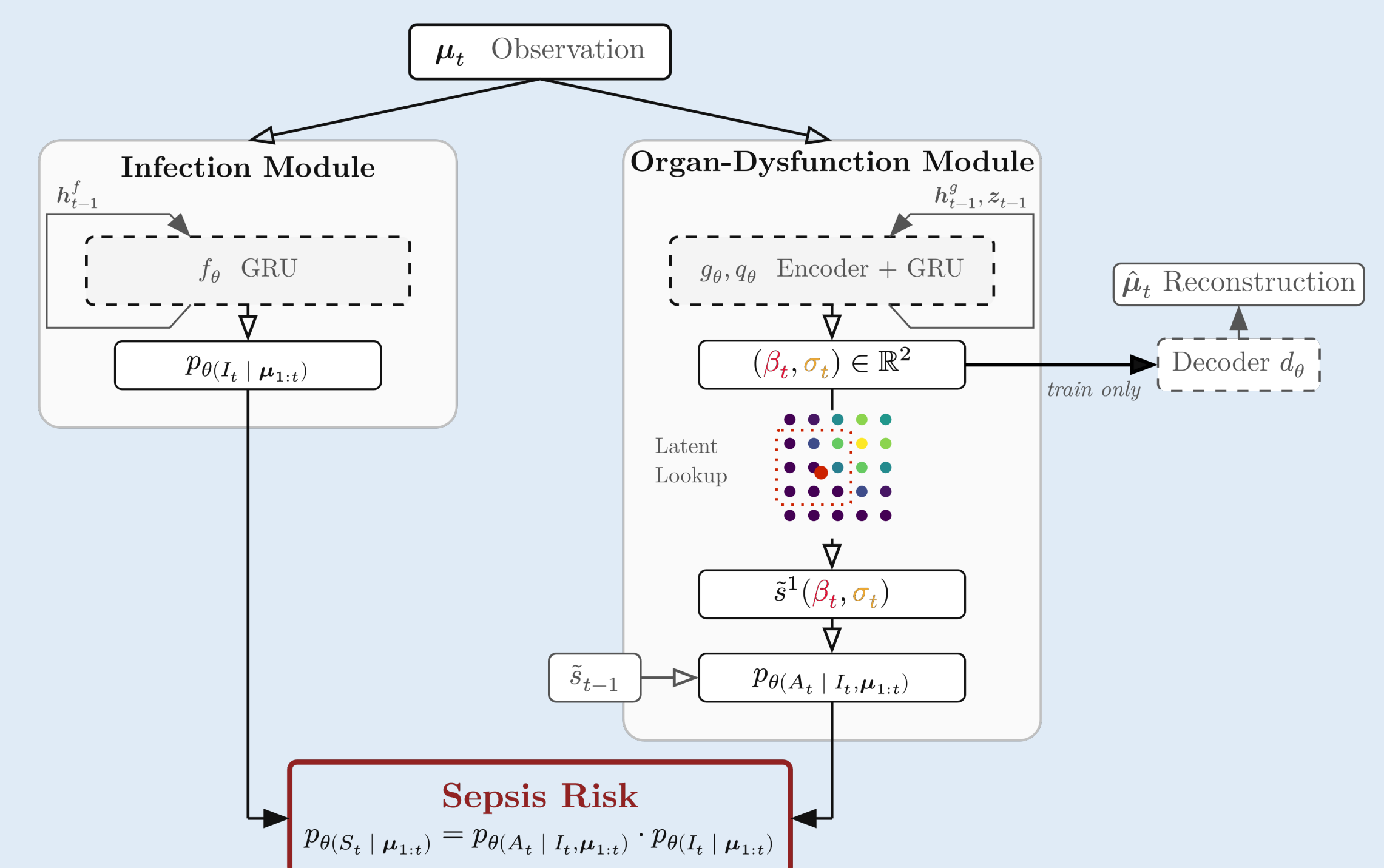


Figure 6: Latent Dynamics Model Architecture

Results

The proposed method is trained and evaluated on real intensive care patients, from a retrospective MIMIC-IV [4] cohort.

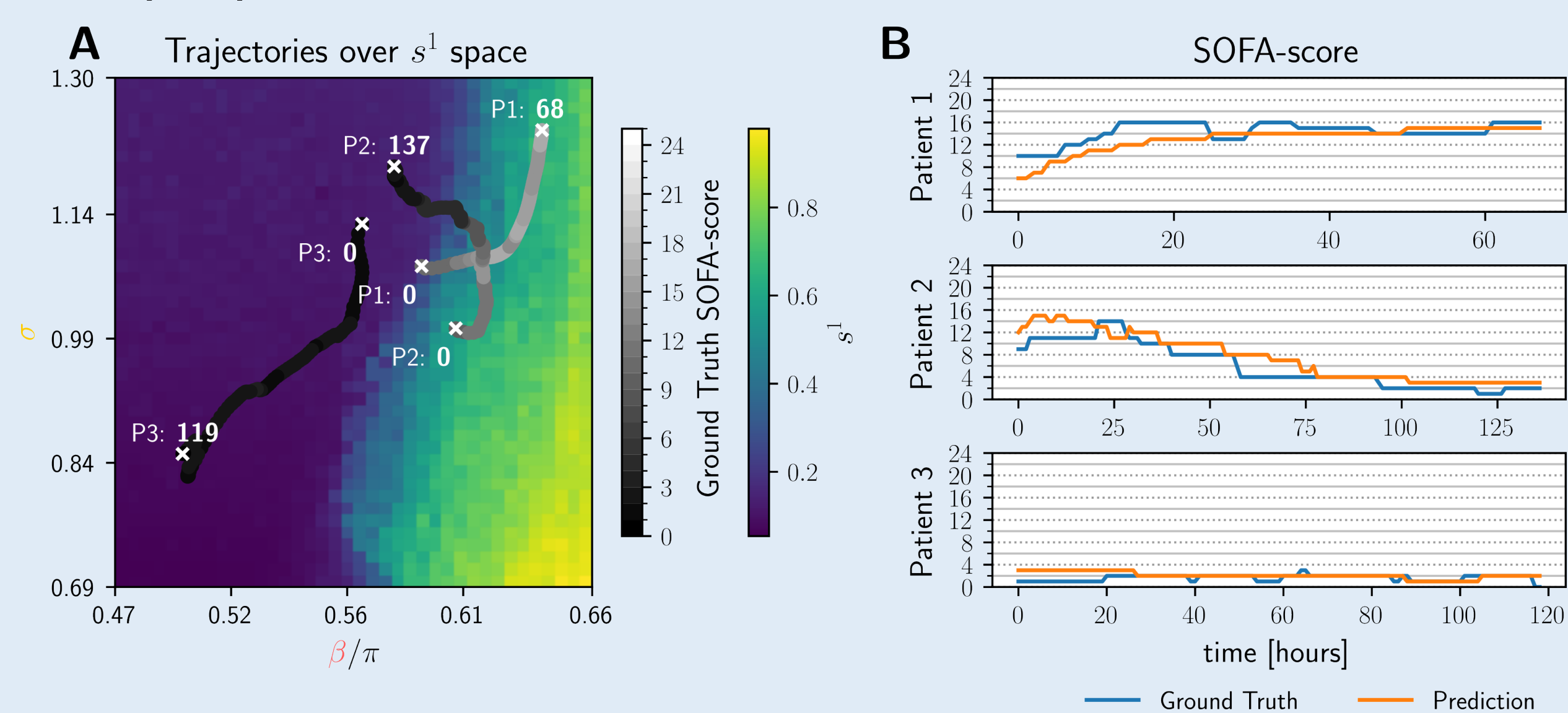


Figure 7: **A** Patient trajectories in (β, σ) space, colored by ground-truth SOFA scores (grayscale). **B** Time series comparing predicted (orange) vs. ground-truth (blue) SOFA scores for the patients.

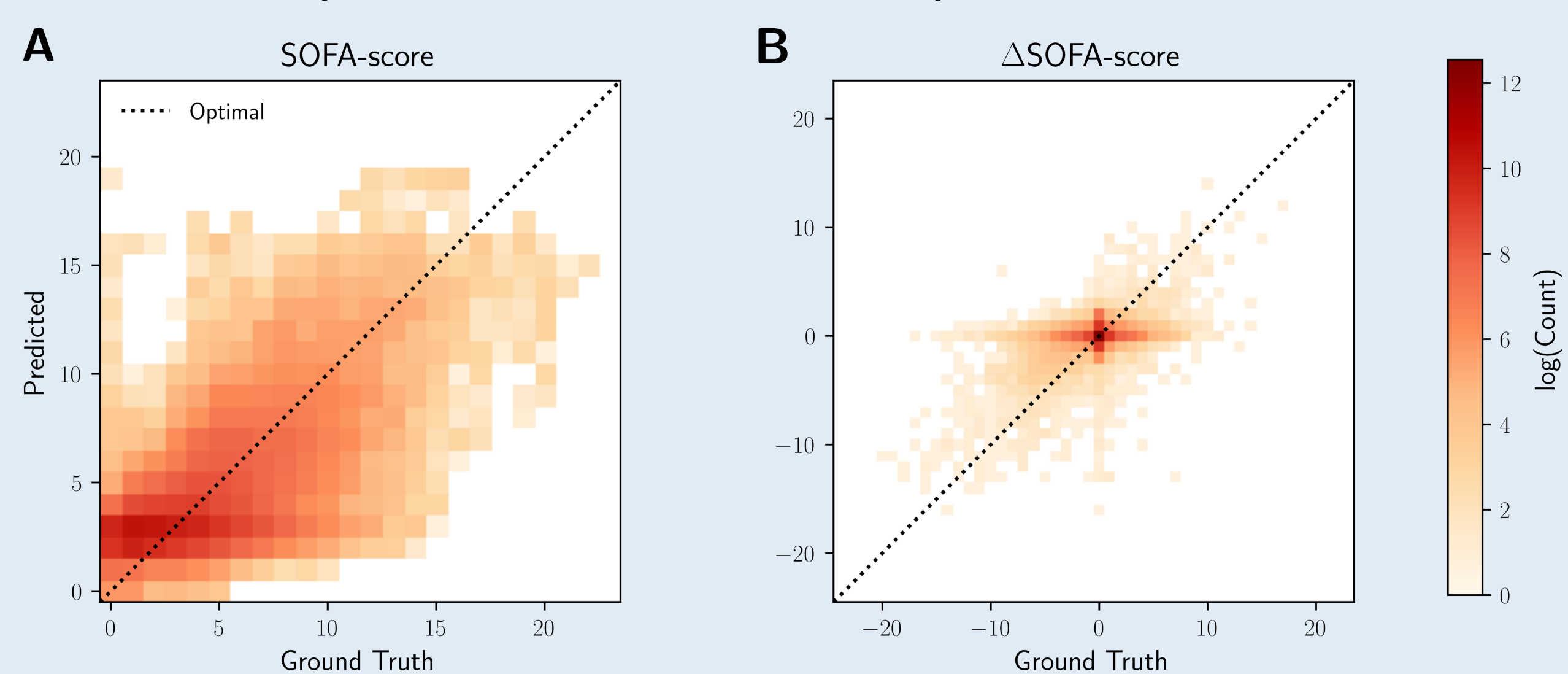


Figure 8: **A** (SOFA) and **B** (Δ SOFA) show the model accurately captures severity and temporal shifts. Log-scale color indicates sample density

The hybrid modeling, combining physiologically grounded dynamics with data-driven learning, can overcome limitations of purely mechanistic and purely black-box approaches to sepsis prediction. Latent trajectories show clinically interpretable patterns of deterioration and recovery, aligned with physiological progression while prediction performance significantly exceeds existing baselines.

¹ J. Sawicki, R. Berner, T. Löser and E. Schöll "Modeling tumor disease and sepsis by networks of adaptively coupled phase oscillators", *Frontiers in Network Physiology*, 2674-0109, 2022.

² R. Berner, J. Sawicki, M. Thiele, T. Löser and E. Schöll "Critical Parameters in Dynamic Network Modeling of Sepsis", *Frontiers in Network Physiology*, 2674-0109, 2022.

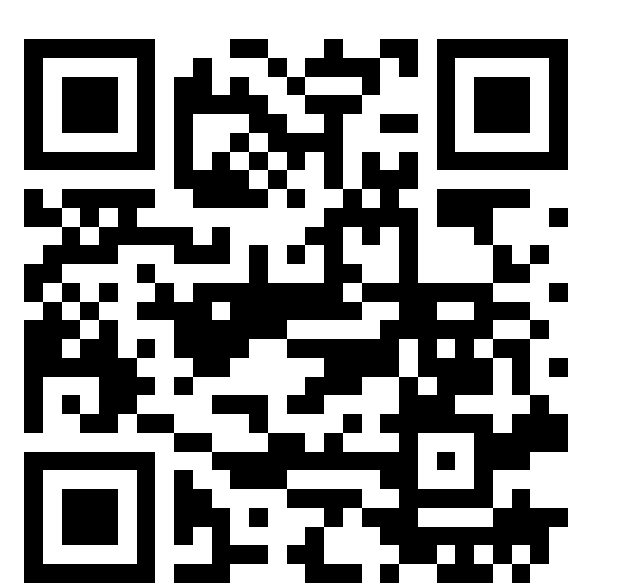
³ M. Singer et al. "The Third International Consensus Definitions for Sepsis and Septic Shock (Sepsis-3)", *JAMA*, 2315:801-810, 2016.

⁴ A. Johnson et al. "MIMIC-III Clinical Database (version 1.4)", *PhysioNet*, RRID:SCR_007345, 2016.



Corresponding Author:
Juri Backes
E-mail:juri.backes@tuhh.de

Repository



https://github.com/unartig/sepsis_osc

- Master Thesis
- Code